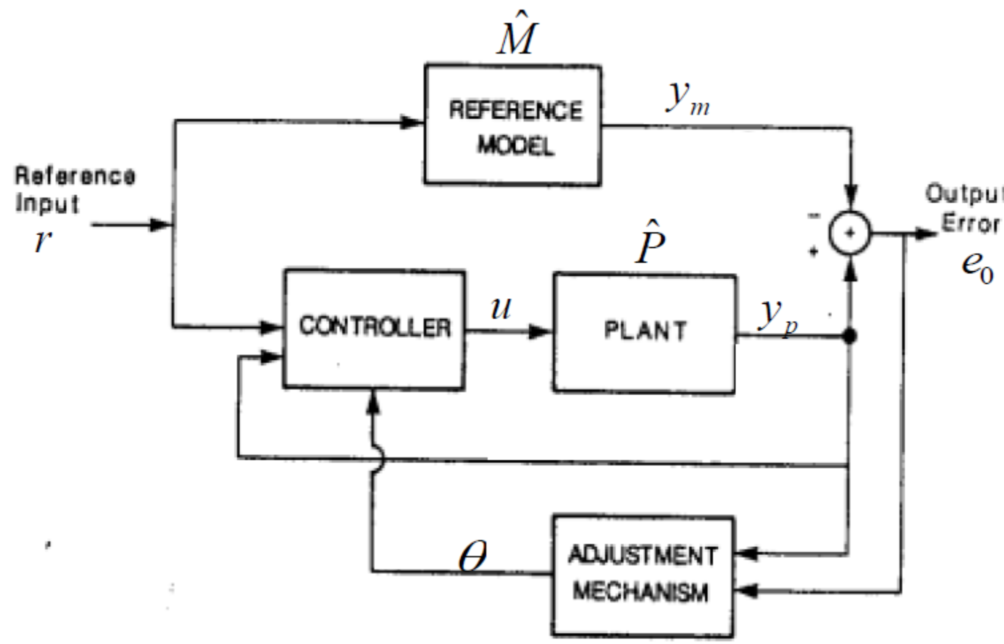
MIT Rule

The MIT Rule

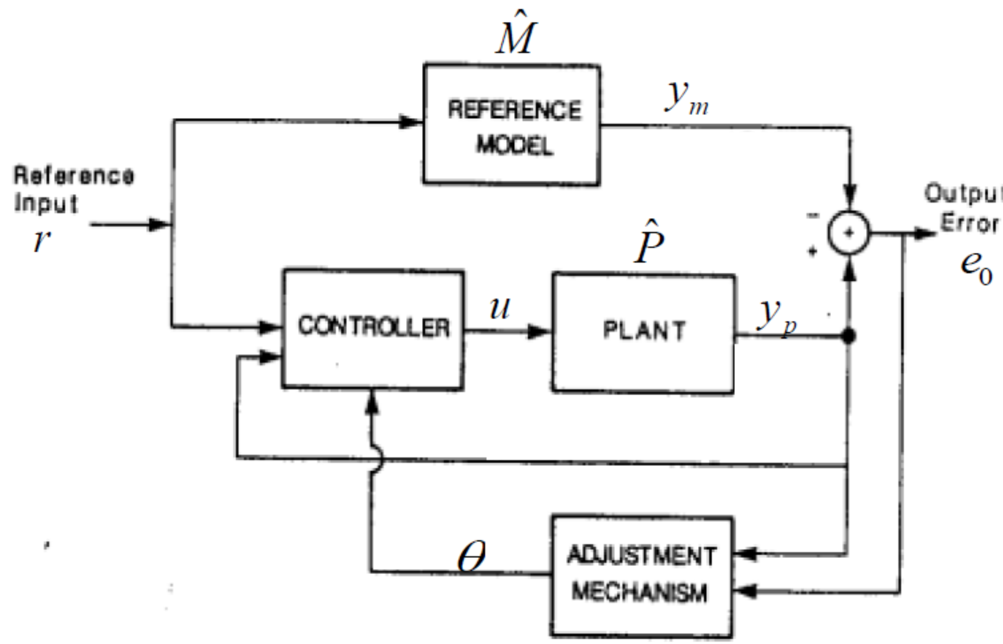
Concept of MRAC



- Design controller to drive plant response to mimic ideal response (error $\rightarrow 0$)
- Designer chooses the reference model, controller structure, and tuning gains for adjustment mechanism

The MIT Rule

MIT rule

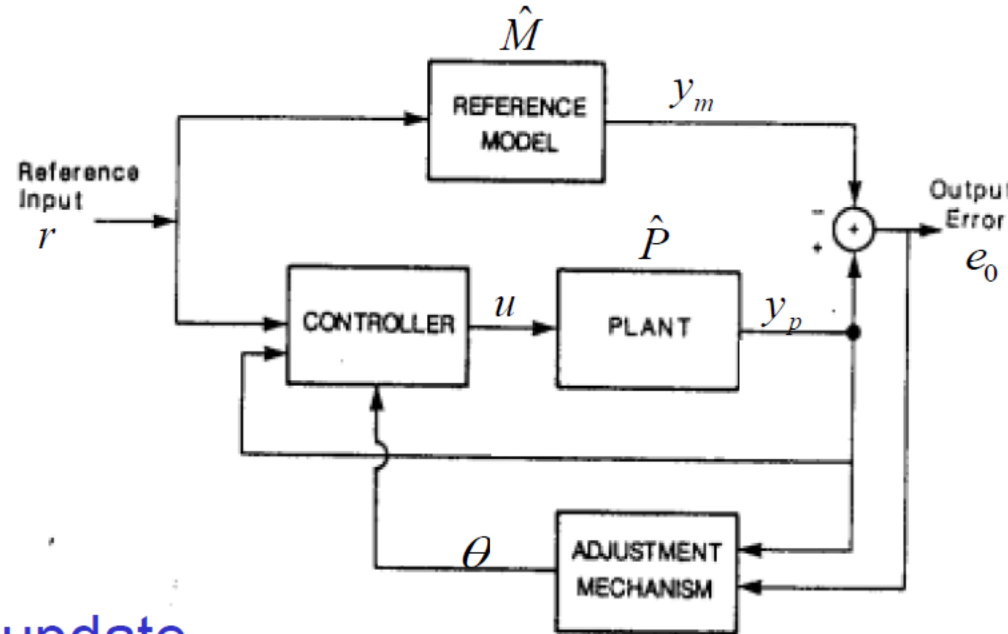


Objective: Adjust the parameter θ to minimize

$$J(\theta) = \frac{1}{2} e_0^2(\theta)$$

The MIT Rule

MIT rule



Gradient update

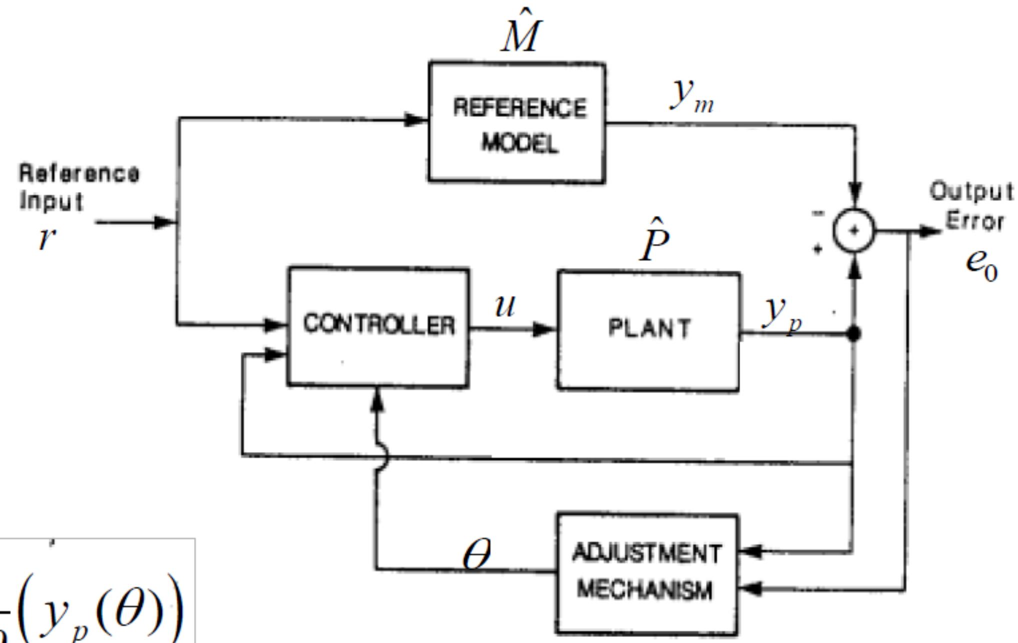
$$\frac{d\theta}{dt} = -\gamma \frac{\partial}{\partial \theta}\left(\frac{1}{2} e_0^2(\theta)\right)$$

Sensitivity function

$$= -\gamma e_0(\theta) \frac{\partial}{\partial \theta}\left(e_0(\theta)\right) = -\gamma e_0(\theta) \frac{\partial}{\partial \theta}\left(y_p(\theta)\right)$$

Adaptation gain (sometimes we use symbol g)

The MIT Rule



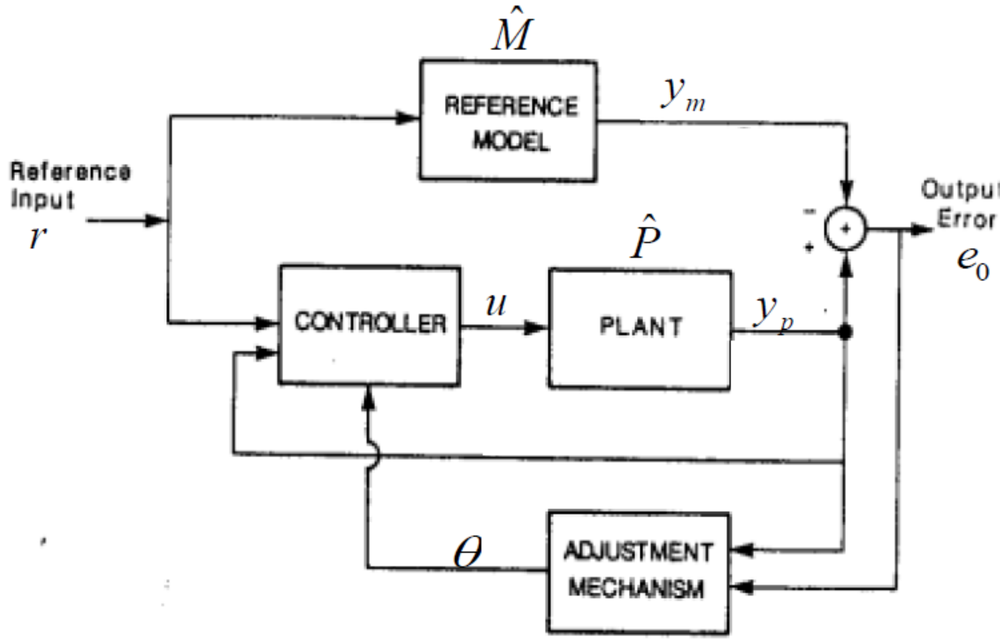
MIT rule

$$\frac{d\theta}{dt} = -\gamma e_0(\theta)\frac{\partial}{\partial\theta}\left(y_p(\theta)\right)$$

MIT rule: an implementation of the gradient update by replacing the unknown parameters in $\partial y_p(\theta)/\partial\theta$ by their estimates at time t

The MIT Rule

MIT rule



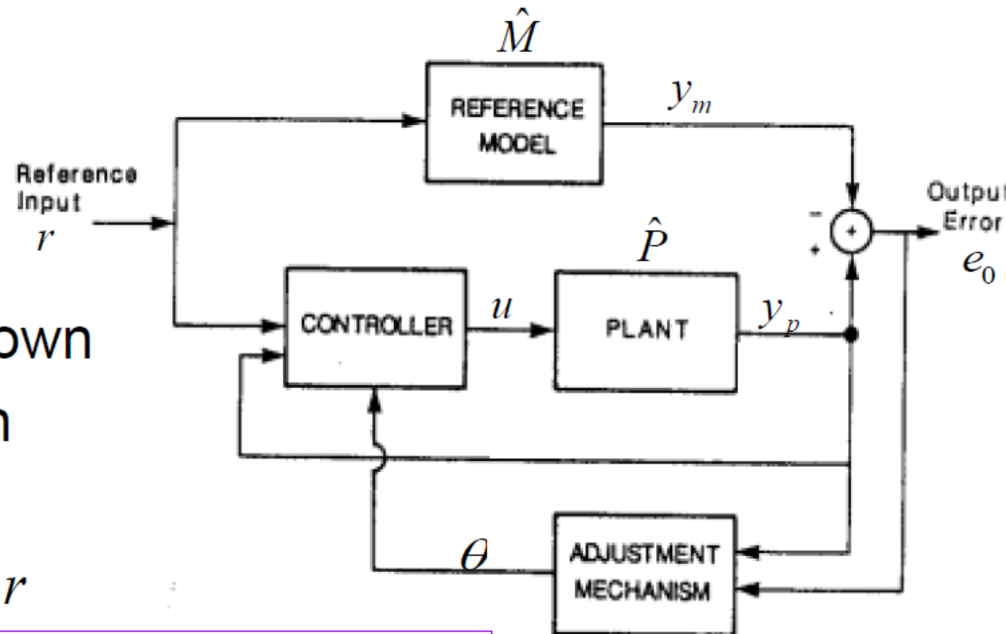
Question: What is the MIT rule if we choose to minimize

$$J(\theta) = |e_0| ?$$

The MIT Rule

Example 1

- Plant: $k\hat{G}(s)$
  where $\hat{G}(s)$ is known
  but k is unknown
- Model: $k_0\hat{G}(s)$
- Controller: $u = \theta r$



Question: $e_0(\theta) = ?$

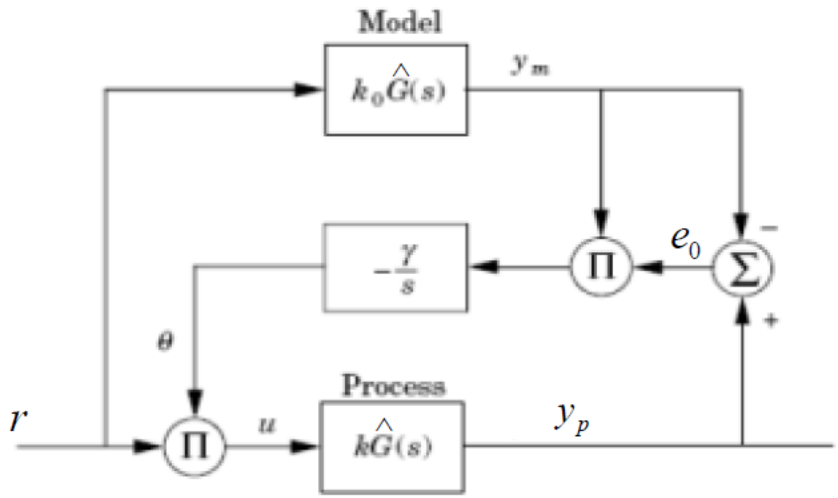
$$\frac{\partial}{\partial \theta}\left(e_0(\theta)\right) = ?$$

$$\frac{d\theta}{dt} = ?$$

The MIT Rule

Example 1

- Plant: $k\hat{G}(s)$
  where $\hat{G}(s)$ is known
  but k is unknown
- Model: $k_0\hat{G}(s)$
- Controller: $u = \theta r$



$$e_0 = y_p - y_m = k\hat{G}(s)\theta r - k_0\hat{G}(s)r$$

$$\frac{\partial e_0}{\partial \theta} = k\hat{G}(s)r = \frac{k}{k_0} y_m$$

$$\frac{d\theta}{dt} = -\gamma' \frac{k}{k_0} y_m e_0 = -\gamma y_m e_0$$

A mixture of time-domain and frequency-domain notations (such hybrid notation is common in the adaptive control literature—it will save us the definition of intermediate variables)

The MIT Rule

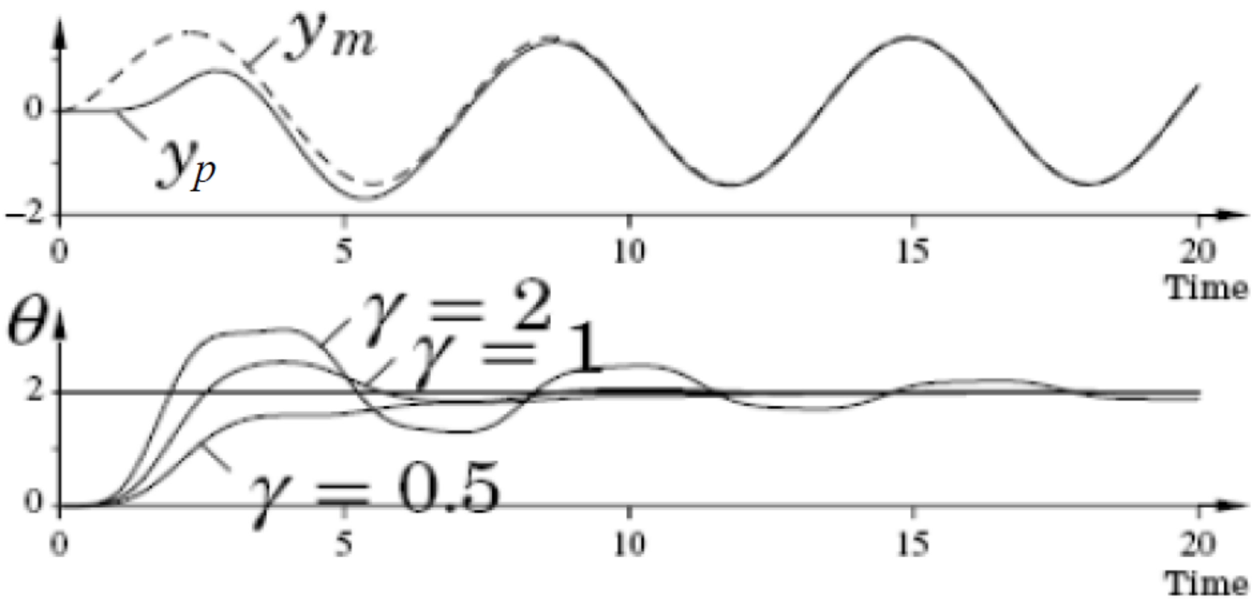
Example 1

$$\hat{G}(s) = \frac{1}{s+1}$$

$$k = 1 \quad \text{and} \quad k_0 = 2$$

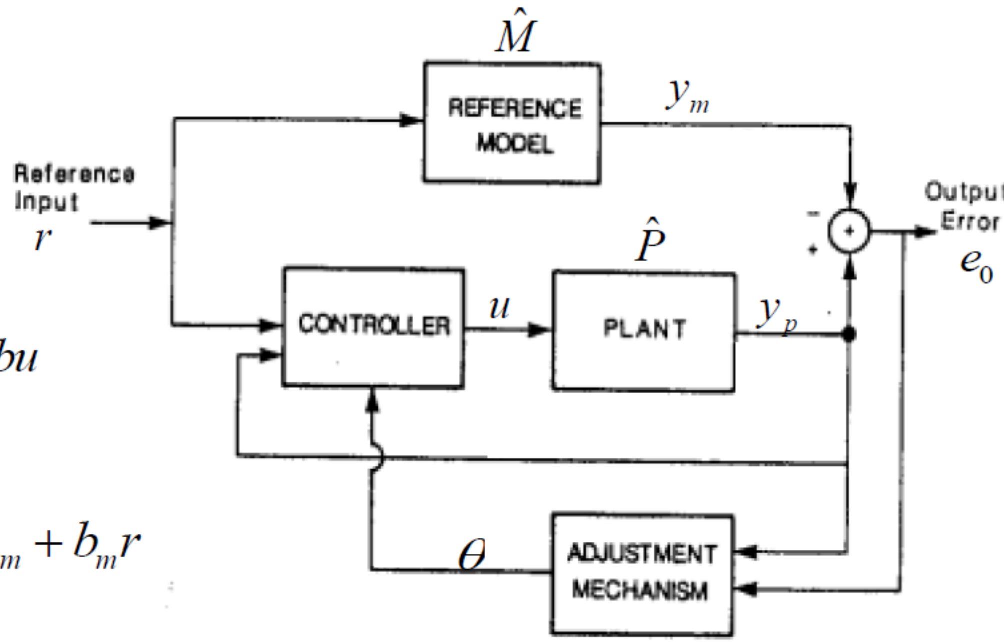
$$r = \sin(t)$$

Simulation



The MIT Rule

Example 2



- Plant:

$$\frac{dy_p}{dt} = -ay_p + bu$$

- Model:

$$\frac{dy_m}{dt} = -a_m y_m + b_m r$$

- Controller: $u = \theta_1 r - \theta_2 y_p$

Question:

$$\frac{\partial}{\partial \theta}\left(e_0(\theta)\right) = ? \qquad \frac{d\theta}{dt} = ?$$

The MIT Rule

Example 2

- Plant: $\frac{dy_p}{dt} = -ay_p + bu$
- Model: $\frac{dy_m}{dt} = -a_m y_m + b_m r$
- Controller: $u = \theta_1 r - \theta_2 y_p$

Question: Can these formulas be used directly in the MIT rule?

$$e_0 = y_p - y_m$$

$$\frac{\partial e_0}{\partial \theta_1} = \frac{b}{s + a + b\theta_2} r$$

$$\frac{\partial e_0}{\partial \theta_2} = -\frac{b^2\theta_1}{(s + a + b\theta_2)^2} r = -\frac{b}{s + a + b\theta_2} y_p$$

The MIT Rule

Example 2

- Plant: $\dfrac{dy_p}{dt} = -ay_p + bu$
- Model: $\dfrac{dy_m}{dt} = -a_m y_m + b_m r$
- Controller: $u = \theta_1 r - \theta_2 y_p$

$$e_0 = y_p - y_m$$

$$\frac{\partial e_0}{\partial \theta_1} = \frac{b}{s + a + b\theta_2} r$$

$$\frac{\partial e_0}{\partial \theta_1} = -\frac{b^2\theta_1}{(s + a + b\theta_2)^2} r = -\frac{b}{s + a + b\theta_2} y_p$$

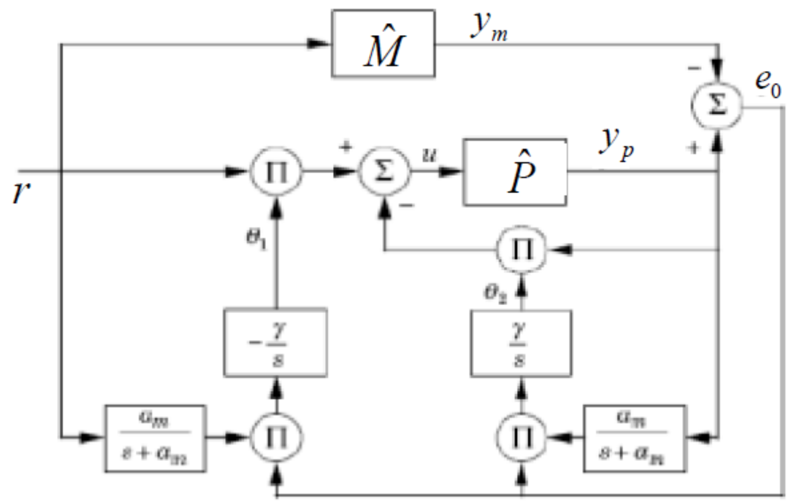
Approximation:

$$s + a + b\theta_2 \approx s + a + b\theta_2^* = s + a_m$$

The MIT Rule

Example 2

- Plant: $\dfrac{dy_p}{dt} = -ay_p + bu$
- Model: $\dfrac{dy_m}{dt} = -a_m y_m + b_m r$
- Controller: $u = \theta_1 r - \theta_2 y_p$



$$e_0 = y_p - y_m$$

$$\frac{\partial e_0}{\partial \theta_1} = \frac{b}{s+a+b\theta_2} r$$

$$\frac{\partial e_0}{\partial \theta_1} = -\frac{b^2\theta_1}{(s+a+b\theta_2)^2} r = -\frac{b}{s+a+b\theta_2} y_p$$

$$s + a + b\theta_2 \approx s + a + b\theta_2^* = s + a_m$$

MIT rule:

$$\frac{d\theta_1}{dt} = -\gamma\left(\frac{a_m}{s+a_m} r\right) e_0$$

$$\frac{d\theta_2}{dt} = \gamma\left(\frac{a_m}{s+a_m} y_p\right) e_0$$

The MIT Rule

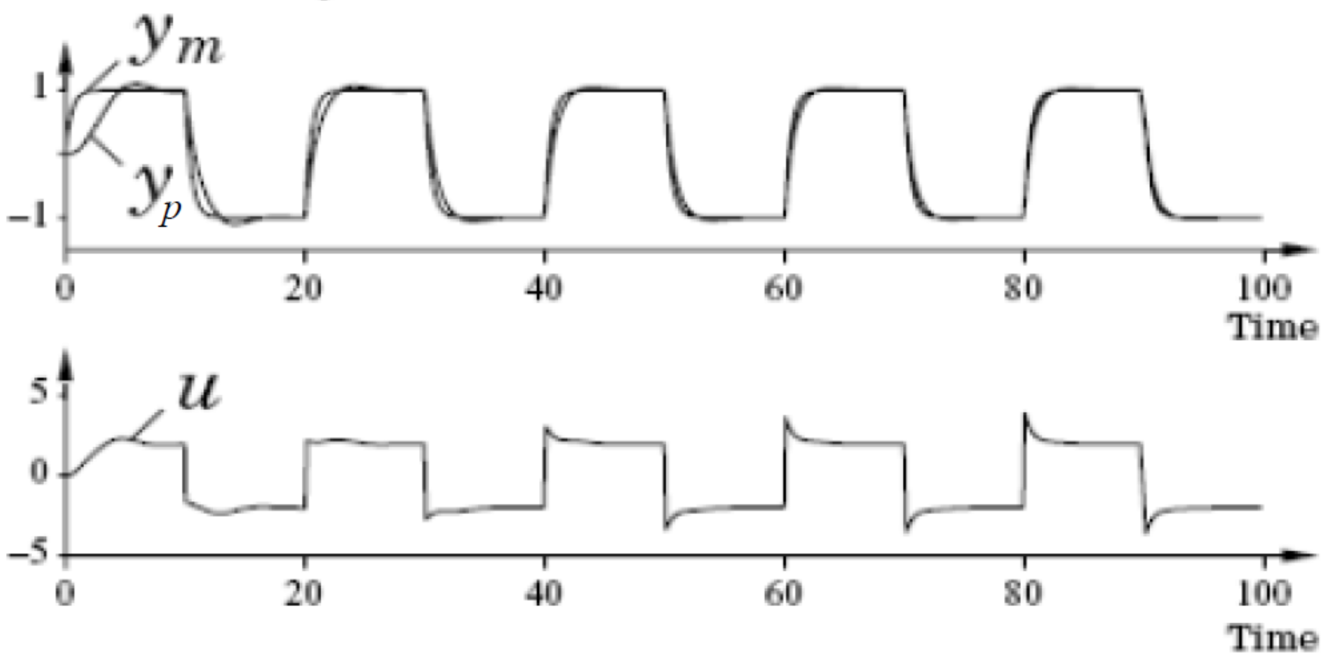
Example 2

$a = 1$ and $b = 0.5$

$a_m = b_m = 2$

$\gamma = 1$

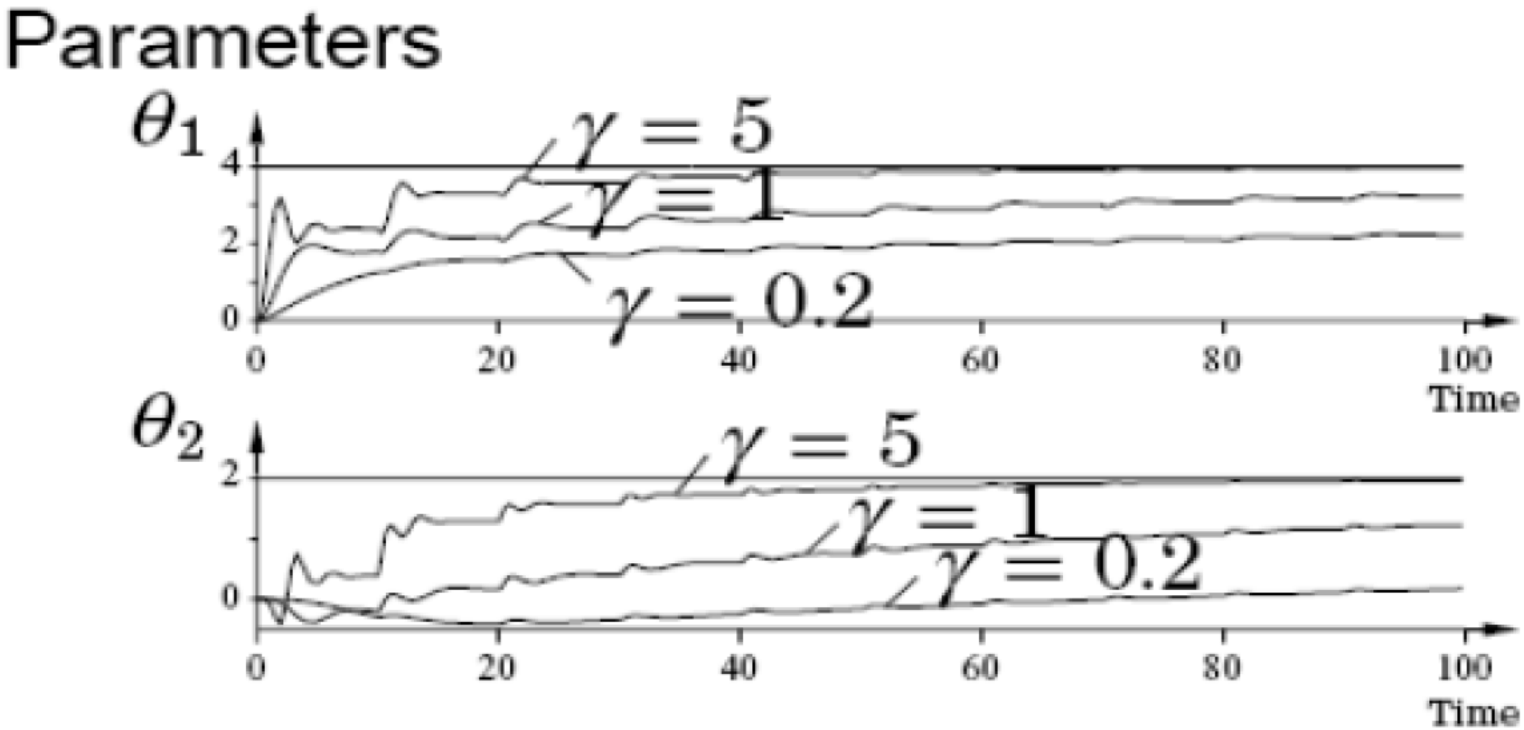
Input and output



The MIT Rule

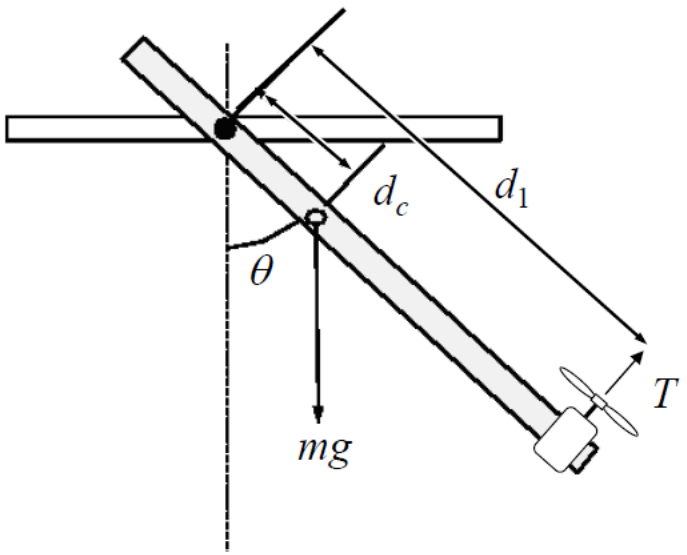
Example 2

$$a = 1 \text{ and } b = 0.5$$
$$a_m = b_m = 2$$
$$\theta_1^* = 4 \text{ and } \theta_2^* = 2$$



The MIT Rule

Example 3



System dynamics

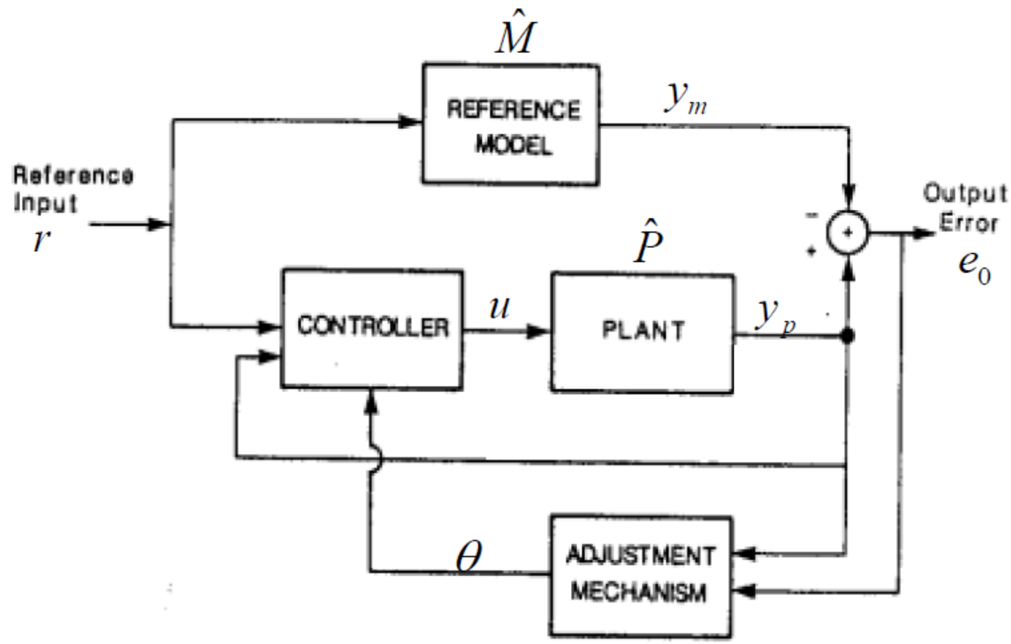
$$J\ddot{\theta} + c\dot{\theta} + mgd_c \sin\theta = d_1 T$$

$$\frac{\theta(s)}{T(s)} = \frac{d_1}{Js^2 + cs + mgd_c}$$

$$\frac{\theta(s)}{T(s)} = \frac{1.89}{s^2 + 0.0389s + 10.77}$$

The MIT Rule

Example 3

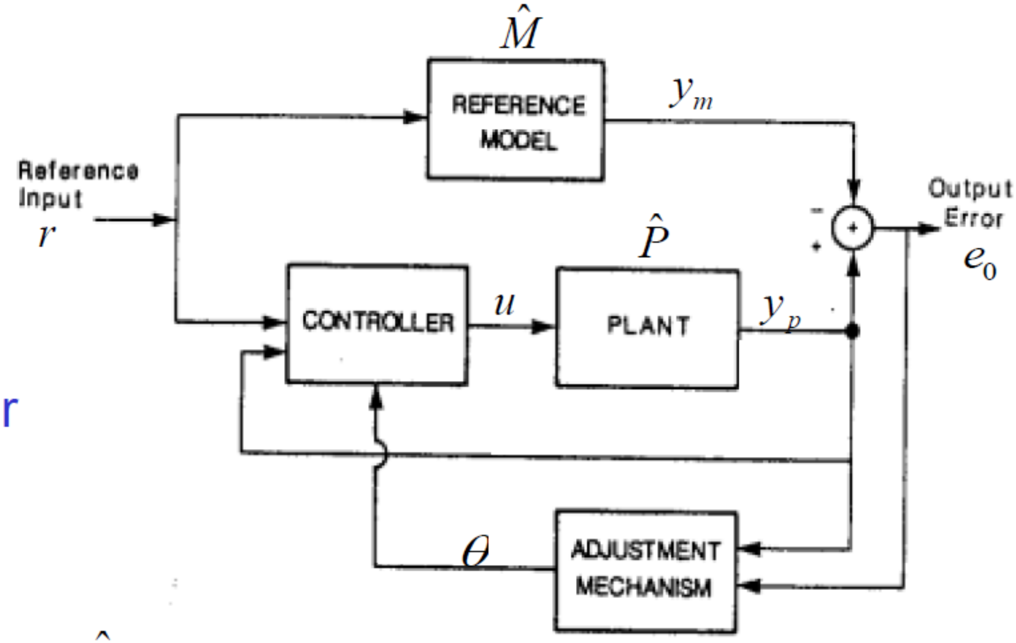


Reference model

$$\widehat{M} = \frac{b_m}{s^2 + a_{1m}s + a_{0m}}$$

The MIT Rule

Example 3



Form of controller

$$u = \theta_1 r - \theta_2 y_p$$

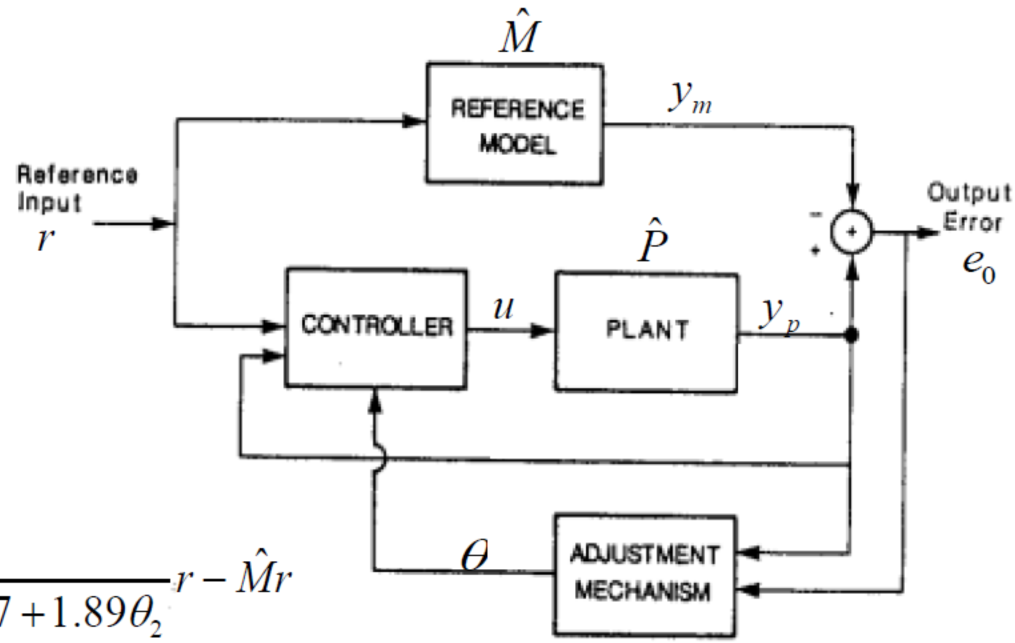
$$e_0 = y_p - y_m = \hat{P}u - \hat{M}r$$

$$y_p = \hat{P}u = \left(\frac{1.89}{s^2 + 0.0389s + 10.77}\right)(\theta_1 r - \theta_2 y_p)$$

$$y_p = \frac{1.89\theta_1}{s^2 + 0.0389s + 10.77 + 1.89\theta_2} r$$

The MIT Rule

Example 3



Gradients

$$e_0 = \frac{1.89\theta_1}{s^2+0.0389s+10.77+1.89\theta_2} r - \hat{M} r$$

$$\frac{\partial e_0}{\partial \theta_1} = \frac{1.89}{s^2+0.0389s+10.77+1.89\theta_2} r$$

$$\frac{\partial e_0}{\partial \theta_2} = -\frac{1.89^2\theta_1}{\left(s^2+0.0389s+10.77+1.89\theta_2\right)^2} r = -\frac{1.89}{s^2+0.0389s+10.77+1.89\theta_2} y_p$$

$$\boxed{y_p = \frac{1.89\theta_1}{s^2+0.0389s+10.77+1.89\theta_2} r}$$

The MIT Rule

Example 3

$$\frac{\partial e_0}{\partial \theta_1} = \frac{1.89}{s^2 + 0.0389s + 10.77 + 1.89\theta_2} r$$

$$\frac{\partial e_0}{\partial \theta_2} = -\frac{1.89}{s^2 + 0.0389s + 10.77 + 1.89\theta_2} y_p$$

Approximation of the gradients

$$y_p = \frac{1.89\theta_1}{s^2 + 0.0389s + 10.77 + 1.89\theta_2} r$$

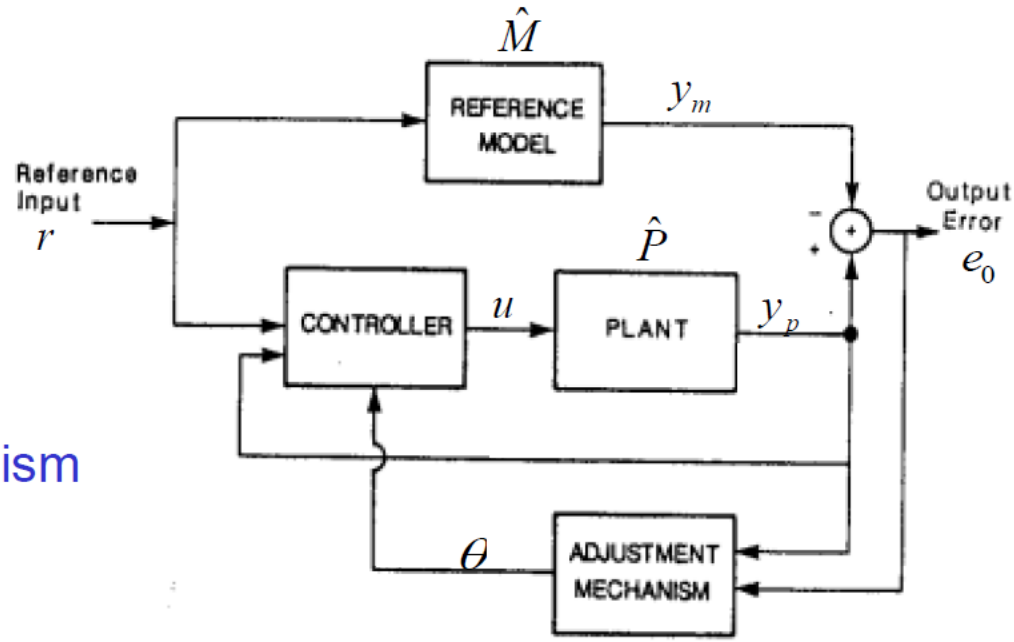
$$s^2 + 0.0389s + 10.77 + 1.89\theta_2 \approx s^2 + a_{1m}s + a_{0m}$$

$$\frac{\partial e_0}{\partial \theta_1} = \frac{1.89}{s^2 + a_{1m}s + a_{0m}} r$$

$$\frac{\partial e_0}{\partial \theta_2} = -\frac{1.89}{s^2 + a_{1m}s + a_{0m}} y_p$$

The MIT Rule

Example 3



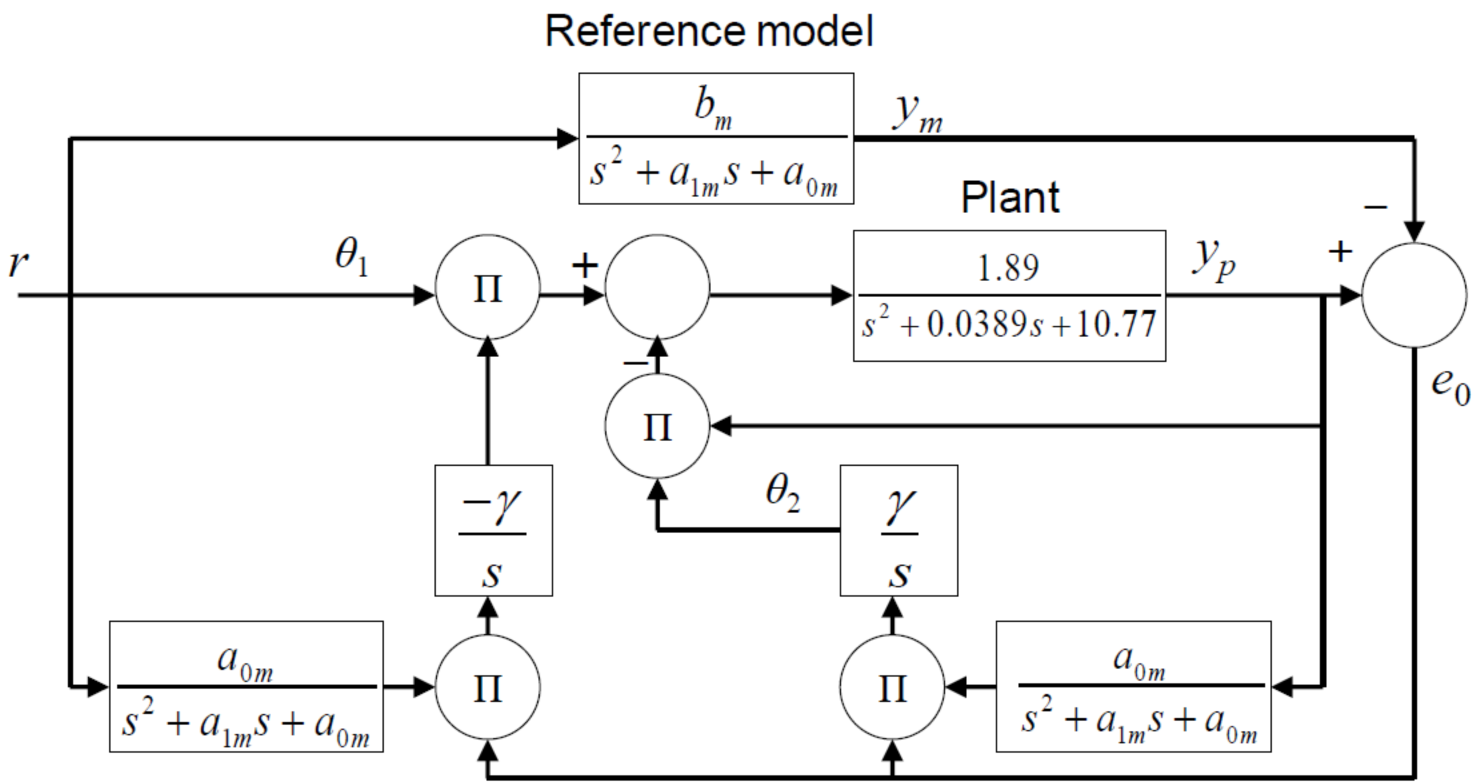
Adjustment mechanism

$$\frac{d\theta_1}{dt} = -\gamma'\frac{\partial e_0}{\partial \theta_1}e_0 = -\gamma\left(\frac{a_{0m}}{s^2 + a_{1m}s + a_{0m}}r\right)e_0$$

$$\frac{d\theta_2}{dt} = -\gamma'\frac{\partial e_0}{\partial \theta_2}e_0 = \gamma\left(\frac{a_{0m}}{s^2 + a_{1m}s + a_{0m}}y_p\right)e_0$$

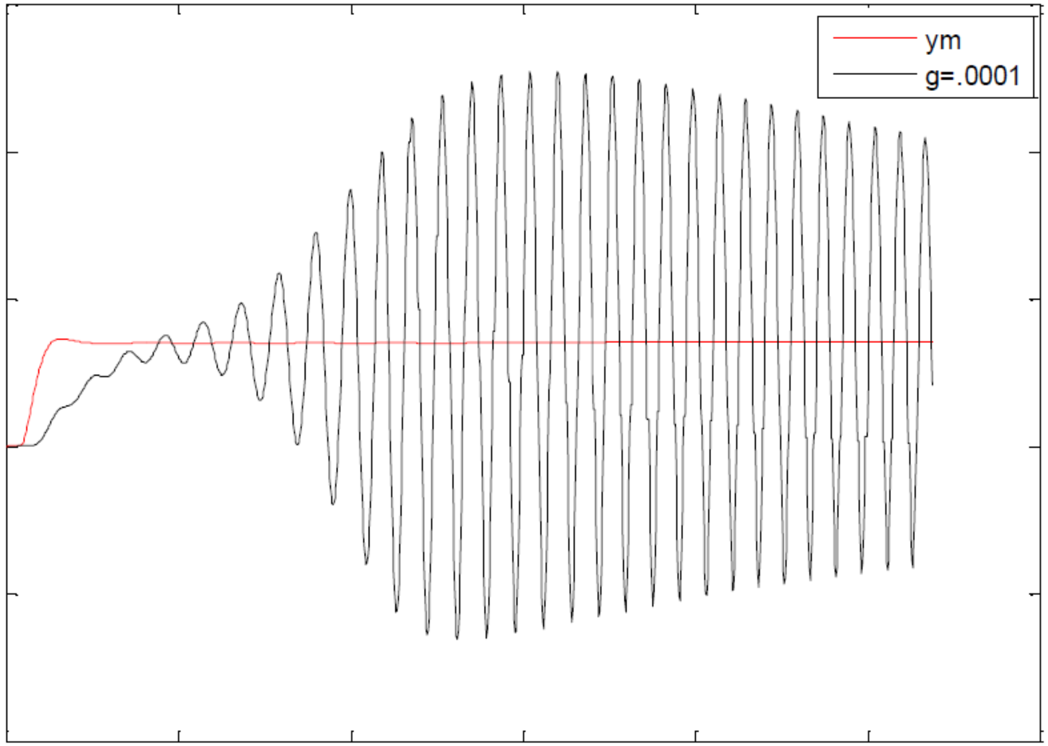
The MIT Rule

Example 3



The MIT Rule

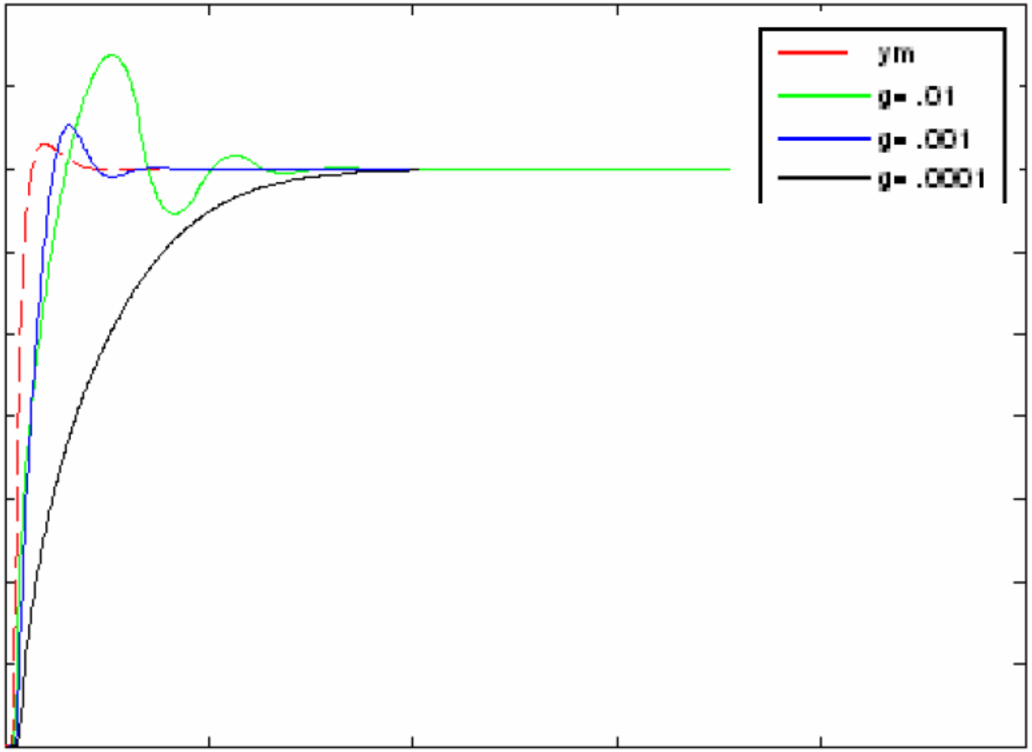
Example 3



Remark: The plant almost goes unstable. This response is largely due to the near instability of the open loop system. Tuning of gamma and changing the reference model did not alleviate this problem.

The MIT Rule

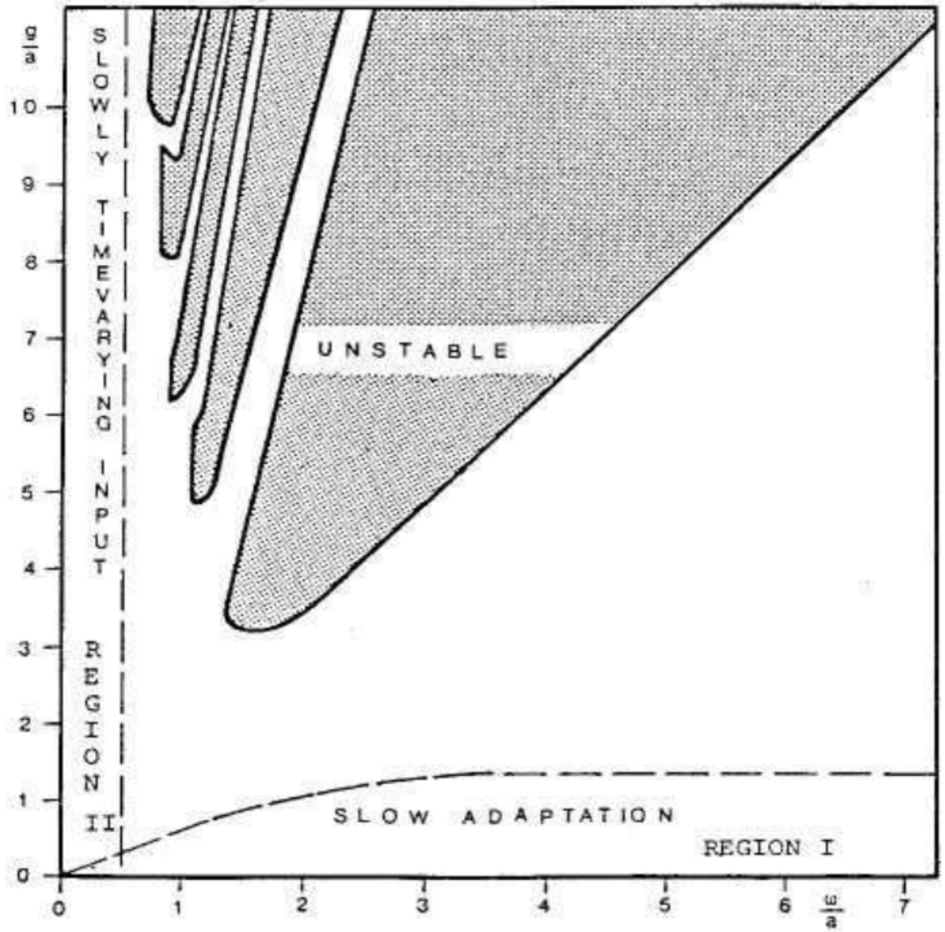
Example 3



Remark: For this specific example, introducing PD control may stabilize the system dynamics.

The MIT Rule

Stability of the MIT rule



$$\hat{P}(s) = \frac{k}{s+1}, \ \hat{M}(s) = \frac{k_0}{s+1}$$

$$r = \sin(\omega t)$$

Remark: In this case, for small enough ω, the MIT rule is stable for all g, while for small enough g, it is stable for all ω. Otherwise, the dynamic behavior of the algorithm is quite complex, with interleaving of the stable and unstable regions.

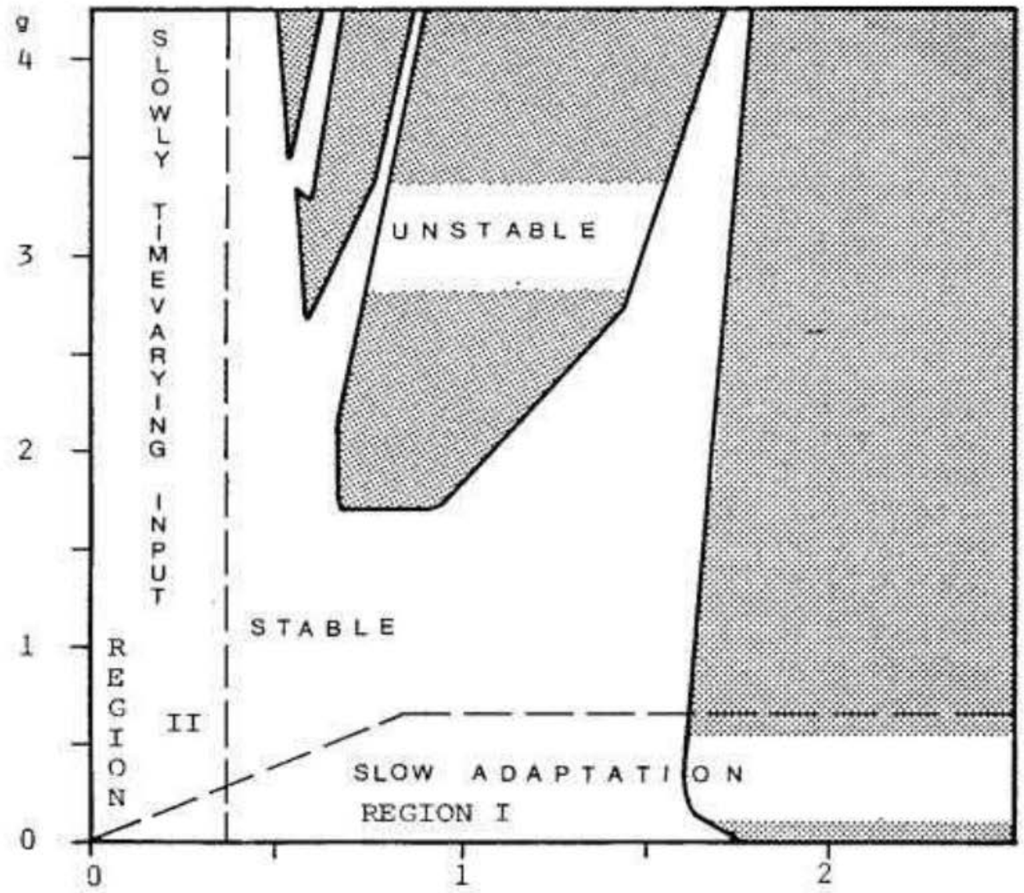
The MIT Rule

Stability of the MIT rule

The case where

$\hat{P}(s)$ and $\hat{M}(s)$

do not share the same set of poles and zeros



The MIT Rule

Tuning the Gain for MIT rule

Consider again the problem with scaling the reference

$$y = k \cdot G(p)\, u, \qquad y_m = k_0 \cdot G(p)\, u_c, \qquad u = \theta\, u_c$$

where $\theta(t)$ is determined by MIT rule:

$$\frac{d}{dt}\theta = -\gamma \cdot y_m \cdot e, \qquad e = y - y_m$$

The equation for θ can be re-written as follows

$$\frac{d}{dt}\theta = -\gamma \cdot y_m \cdot \Big(y - y_m\Big) = -\gamma \cdot y_m \cdot \Big(k \cdot G(p)\,\theta\, u_c - y_m\Big)$$

The MIT Rule

Tuning the Gain for MIT rule

Consider again the problem with scaling the reference

$$y = k \cdot G(p)\, u, \qquad y_m = k_0 \cdot G(p)\, u_c, \qquad u = \theta\, u_c$$

where $\theta(t)$ is determined by MIT rule:

$$\frac{d}{dt}\theta = -\gamma \cdot y_m \cdot e, \qquad e = y - y_m$$

The equation for θ can be re-written as follows

$$\frac{d}{dt}\theta(t) + \gamma \cdot k \cdot y_m(t) \cdot G(p)\Big[\theta(t) u_c(t)\Big] = \gamma y_m^2(t)$$

The MIT Rule

Tuning the Gain for MIT rule

Consider again the problem with scaling the reference

$$y = k \cdot G(p)\, u, \qquad y_m = k_0 \cdot G(p)\, u_c, \qquad u = \theta\, u_c$$

where $\theta(t)$ is determined by MIT rule:

$$\frac{d}{dt}\theta = -\gamma \cdot y_m \cdot e, \qquad e = y - y_m$$

The equation for θ can be re-written as follows

$$\frac{d}{dt}\theta(t) + \gamma \cdot k \cdot y_m(t) \cdot G(p)\Big[\theta(t) u_c(t)\Big] = \gamma y_m^2(t)$$

Here

- the functions $y_m(t)$ and $u_c(t)$ are known!
- the range of the constant gain γ, for which the nominal value θ^0 (its stationary point) is stable, should be determined.

The MIT Rule

Tuning the Gain for MIT rule (cont'd)

$$\frac{d}{dt}\theta(t) + \gamma \cdot k \cdot y_m(t) \cdot G(p)\Big[\theta(t)\, u_c(t)\Big] = \gamma y_m^2(t)$$

In general the analysis of stability is difficult!

Consider the case when $y_m(t) \equiv y_m^o$, $u_c(t) = u_c^o$, then ODE

$$\frac{d}{dt}\theta(t) + \gamma \cdot k \cdot y_m^o \cdot u_c^0 \cdot G(p)\Big[\theta(t)\Big] = \gamma (y_m^o)^2$$

is linear and time-invariant!

Stability is determined by the roots of the algebraic equation

$$s + \mu \cdot G(s) = 0, \qquad \mu = \gamma \cdot k \cdot y_m^o \cdot u_c^0$$

The MIT Rule

Tuning the Gain for MIT rule (cont'd)

$$\frac{d}{dt}\theta(t) + \gamma \cdot k \cdot y_m(t) \cdot G(p)\Big[\theta(t)\, u_c(t)\Big] = \gamma y_m^2(t)$$

In general the analysis of stability is difficult!

Consider the case when $y_m(t) \equiv y_m^o$, $u_c(t) = u_c^o$, then ODE

$$\frac{d}{dt}\theta(t) + \gamma \cdot k \cdot y_m^o \cdot u_c^0 \cdot G(p)\Big[\theta(t)\Big] = \gamma (y_m^o)^2$$

is linear and time-invariant!

Stability is determined by the roots of the algebraic equation

$$s + \mu \cdot G(s) = 0, \qquad \mu = \gamma \cdot k \cdot y_m^o \cdot u_c^0$$

Root locus analysis (variation of zeros with μ) can be used.
A reasonable value for γ can be obtained from this analysis and might work for slowly varying signals.

The MIT Rule

Example 4

Let, as in Example 5.1

$$G(s) = \frac{1}{s+1}, \qquad k = 1, \qquad k_0 = 2$$

The characteristic equation

$$s + \mu \frac{1}{s+1} = 0 \quad \Leftrightarrow \quad s^2 + s + \mu = 0$$

has stable zeros if and only if

$$\mu = \gamma \cdot k \cdot y_m^o \cdot u_c^0 = \gamma \cdot \left(k_0\, G(0)\, u_c^0 \right) \cdot u_c^0 = 2\,\gamma\,(u_c^0)^2 > 0$$

The MIT Rule

Example 5

Consider the stable system with relative degree 2:

$$G(s) = \frac{1}{s^2 + a_1\, s + a_2}, \qquad a_1 > 0, \qquad a_2 > 0$$

The characteristic equation

$$s + \mu \frac{1}{s^2 + a_1\, s + a_2} = 0 \quad \Leftrightarrow \quad s^3 + a_1\, s^2 + a_2\, s + \mu = 0$$

has stable zeros if and only if

$$\mu > 0 \qquad \text{and} \qquad a_1\, a_2 > \mu = \gamma \cdot k \cdot y_m^o \cdot u_c^0$$

Conclusion: with any choice of $\gamma > 0$, stability is lost for sufficiently large magnitudes of the reference signal u_c^0!

The MIT Rule

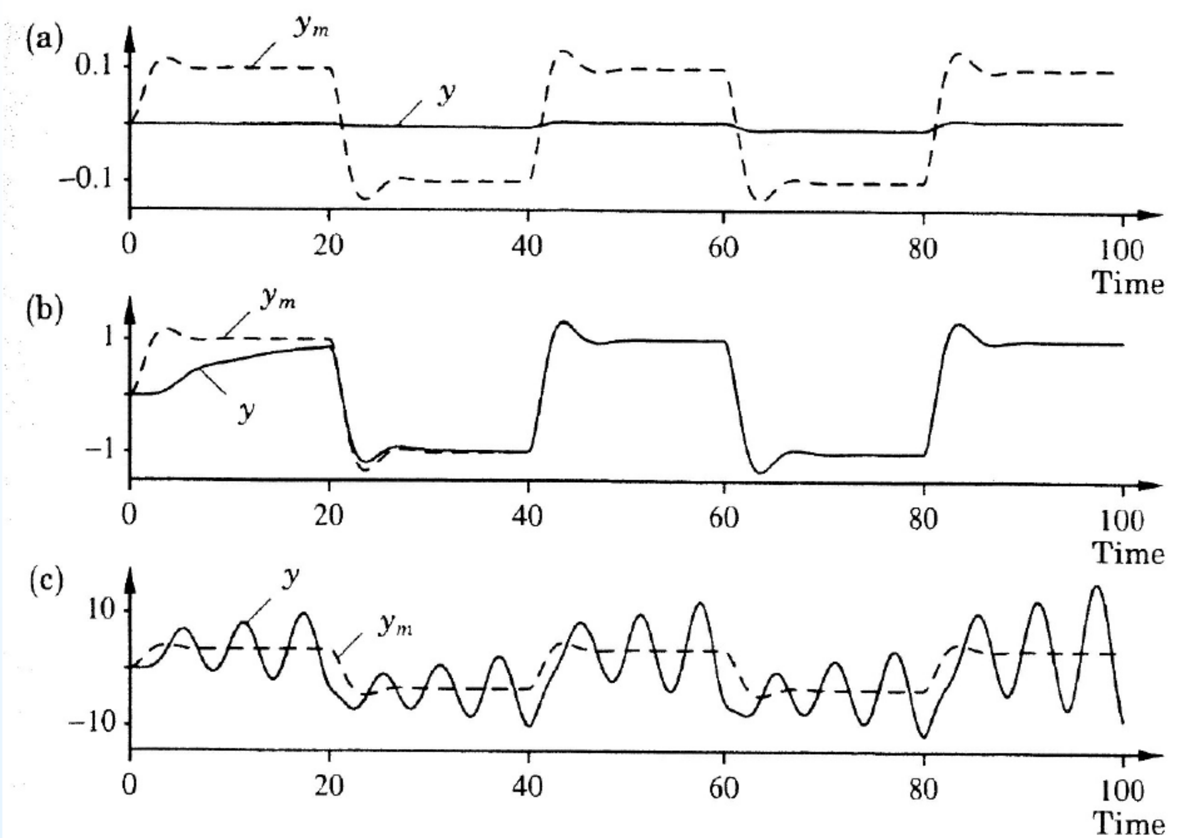


Figure 5.8 Simulation of the MRAS in Example 5.5. The command signal is a square wave with the amplitude (a) 0.1, (b) 1, and (c) 3.5. The model output y_m is a dashed line; the process output is a solid line. The following parameters are used: $k = a_1 = a_2 = \theta^0 = 1$, and $\gamma = 0.1$.

The MIT Rule

Normalized MIT rule

$$\frac{d}{dt}\theta = -\gamma \cdot e(t,\theta) \cdot \frac{\phi}{\alpha + \phi^T \phi}, \qquad \phi = \frac{\partial}{\partial \theta} e(t,\theta), \qquad \alpha > 0$$

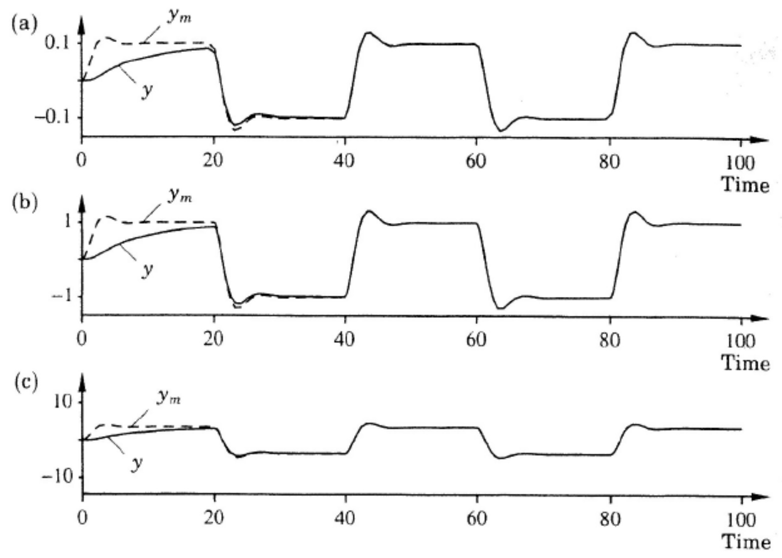


Figure 5.9 Simulation of the MRAS in Example 5.5 with the normalized MIT rule. The command signal is a square wave with the amplitude (a) 0.1, (b) 1, and (c) 3.5. Compare with Fig. 5.8. The model output y_m is a dashed line; the process output is a solid line. The parameters used are $k = a_1 = a_2 = \theta^0 = 1$, $\alpha = 0.001$, and $\gamma = 0.1$.